As is shown in Figure S.1 (center), the spectrum of the singular values peaked around zero but widely spread up to its edge. Figure S.1 (right) demonstrates that the largest singular value becomes monotonically large for the increase of a. Because the larger singular value implies the lower trainability (Bjorck et al., 2018), this is consistent with the empirical observation of (Golubeva et al., 2021) and our Figure S.1 (left).

In contrast, the Mixers are unlikely to suffer from the large singular values as follows. Suppose S-Mixer with $S = C \gg 1$ for simplicity. Then, each layer of the effective MLP has $p = 1/C$ which corresponds to the scaling index $a = 1/3$ in SW-MLP. Actually, its singular value becomes further better than $a = 1/3$, because the weight matrices of the normal and RP Mixers are structured: Consider the singular values of $Z = J_2(I_C \otimes W)J_1$ with a $C \times C$ random Gaussian matrix W and permutation matrices (J_1, J_2). Then, the singular values of Z are equivalent to those of W, excluding duplication. Therefore, the singular values of the Mixers are determined only by the dense weight matrix W. Define $Q = WW^\top$. Because the normalized matrix Q/c obeys the Marchenko-Pastur law in the random matrix theory and its largest eigenvalue is given by 4 in the infinite C limit (Bai & Silverstein, 2010). This means that the largest singular value of the normalized W is 2 and corresponds to $a = 0$ of SW-MLP (i.e., dense MLP) with the infinite Ω limit in Figure S.1 (right). Thus, we can say that from the perspective of random weights, the trainability of the Mixers is expected to be better than that of SW-MLP.

We can also extend our analysis to the models incorporating the expansion factor γ. For SW-MLP with MLP blocks, the expected number of weights is given by $\Omega = \gamma p m^2$. We just need to replace p in the S-Mixer case to γp and the monotonic increase of the largest singular value appears as well. For the MLP-Mixer, its normalized W is a $\gamma C \times C$ matrix. According to the Marchenko-Pastur law for rectangular random matrices, as $C \to \infty$, the largest singular value approaches a constant value of $1 + \sqrt{\gamma}$. This corresponds to the singular value of $a = 0$ in the corresponding SW-MLP, and the result is similar as in the S-Mixer.

Note that the effective width of the mixing layers is sufficiently large but still has an upper bound (15). It satisfies

$$(\sqrt{1 + 8\Omega/\gamma} - 1)/2 \leq m \leq (\Omega/\gamma)^{2/3}, \tag{S.45}$$

where the equality of the lower bound holds for $S = 1$ or $C = 1$. In contrast, for SW-MLP, we have no upper bound and only the lower bound $\sqrt{\Omega} \leq m$, where this equality holds for a dense layer. We can consider an arbitrarily small p and a large m for a fixed Ω if we neglect the issue of memory (Golubeva et al., 2021). Golubeva et al. (2021) reported that extremely small p can cause a decrease in test accuracy owing to the deterioration of trainability. We observed a similar deterioration for the SW-MLP, as shown in this section, but not for the Mixers. This is expected because m is upper-bounded in the Mixers and the trainability is less damaged than that of the SW-MLP with high sparsity.

D.2. MLP-Mixer with increasing width

Figure S.2 shows the test accuracy improves as the width increases on the models SW-MLP and normal and RP MLP-Mixer even if the expansion factor γ is incorporated in the models. We set the expansion factor $\gamma = 4$. We set the initial learning rate to be 0.1. For normal and RP MLP-Mixer, we set $C = 16, 32, 48, 64, 83, 113, 173$ and determined S by combinations of C and $\Omega = 2^{18}, 2^{20}$. For SW-MLP, we set $p = 0.1, 0.3, 0.5, 0.7, 0.9$ and set the width by $m = \sqrt{\Omega/\gamma p}$.

D.3. Increasing expanding factor

The effective MLP expression of the MLP-Mixer (S.21) has two widths: $m = SC$ and γSC. As both are proportional to SC, we have focused on changing SC and fixed γ so far. Here, we consider a complementary setting, that is, changing γ with fixed $m = SC$. By substituting $S = m/C$ into (14), we obtain $\gamma = 2\Omega/(m(C + m/C))$. Similar to m of C shown in Fig. 3, this γ is a single-peak function of C and takes its maximum as

$$C^* = S^* = \sqrt{m}, \quad \max_{S,C} \gamma = \Omega/(m\sqrt{m}). \tag{S.46}$$

Fig. S.3 (left) confirms that increasing the width (γ) leads to performance improvement as is expected from Hypothesis4.1. We trained normal and RP MLP-Mixers with various γ in a realistic range. We plotted some cases of fixed Ω under the same m. Fig. S.3 (right) shows the test accuracy maximized around $C = S$ as is expected from (S.46).

D.4. Dependence on depth

As shown in Figs.5-S.3 , both the normal and RP Mixers exhibited similar tendencies for a fixed depth. Fig. 7 confirms that by increasing the depth, i.e., the number of blocks, RP S-Mixers can even become comparable to the normal ones or

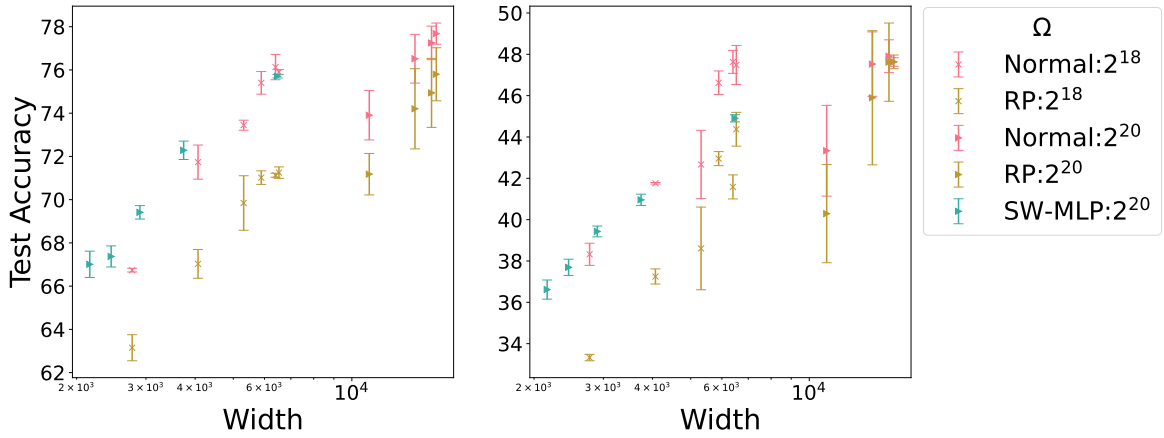

Figure S.2 . Test accuracy improves as the effective width increases. MLP-Mixer, RP-MLP-Mixer, and SW-MLP with $\gamma = 4$ on (Left) CIFAR-10, (Right) CIFAR-100.

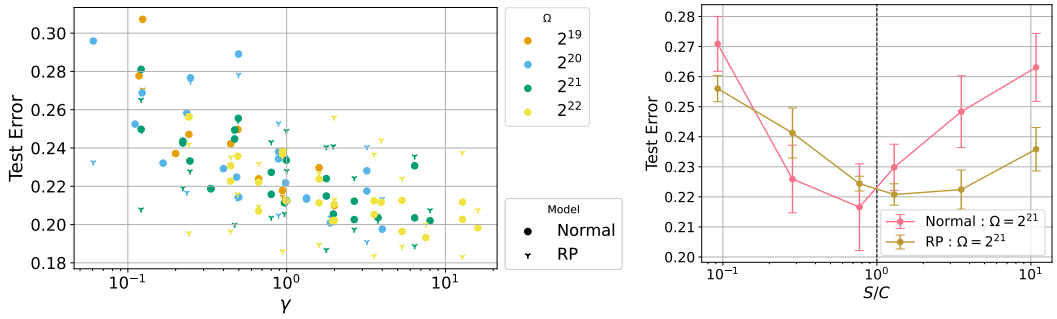

Figure S.3 . (Left) Increasing expansion factor γ improved the test error in normal and RP MLP-Mixers. (Right) The lowest error is achieved around $C = S$ with fixed $m = 4096$.

better than them in some cases. First, we observed that, when the depth was limited, the RP Mixers were inferior to the normal Mixers in most cases. As we increased the depth, we observed that in some cases, overfitting occurred for the normal Mixer, but not for the RP one (see also the training and test losses shown in Section D.6). In such cases, the results of the RP Mixers were comparable (in Figs. 7(left, right)) or better (in Fig. 7(center)). Although RP Mixers are not necessarily better than normal ones, it is intriguing that even RP Mixers defined by a random structure can compete with normal Mixers.

The random permutation in RP-Mixer initially selects and fixes *the size of receptive field* (i.e., the number of input pixels flowing into a neuron in each upper layer through weights) at a rate of $1/\sqrt{m}$. Therefore, the training becomes difficult in shallow networks. However, in the RP-Mixer, since an independent random permutation is chosen for each layer, the receptive field expands as the number of layers increases. From this, the performance of the RP-Mixer improves as layers are added. The normal MLP-Mixer performs well even with fewer layers because the receptive field and diagonal-block structure are aligned.

D.5. Replacement of J_c in specific layers

Figure S.4 provides more detailed insight into the case where the depth is limited and RP Mixers perform worse than normal Mixers. We investigated special S-Mixers whose l-th block was replaced with its RP counterpart while the other layers remained the same. Interestingly, when the accuracy deterioration apparently appears (e.g., cases of CIFAR-10 and STL-10 in Figure 7), this deterioration is attributed to the first block. This seems rational because the neighboring image patches are likely to be correlated, which makes the input units to the first token mixing correlated. Although the usual mixing weights can reflect such neighboring structures, RP Mixers randomly choose tokens and may lose the neighboring structure specific to the images. However, as the depth increases, the token mixing layers can merge all the tokens, which is consistent with the increase in the accuracy of the RP Mixers, as confirmed in Figure 7. Thus, we conclude that the RP and normal mixing

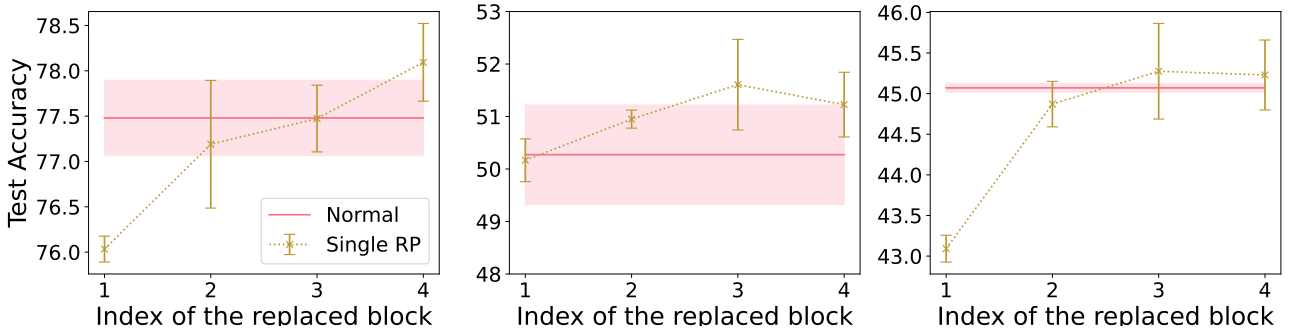

Figure S.4 . Replacing a single block of the normal Mixer with a corresponding RP block clarifies that the upstream layers are functionally commutative with the RP block. (Left) CIFAR-10, (Center) CIFAR-100, (Right) STL-10. We set $C = S = 128$.

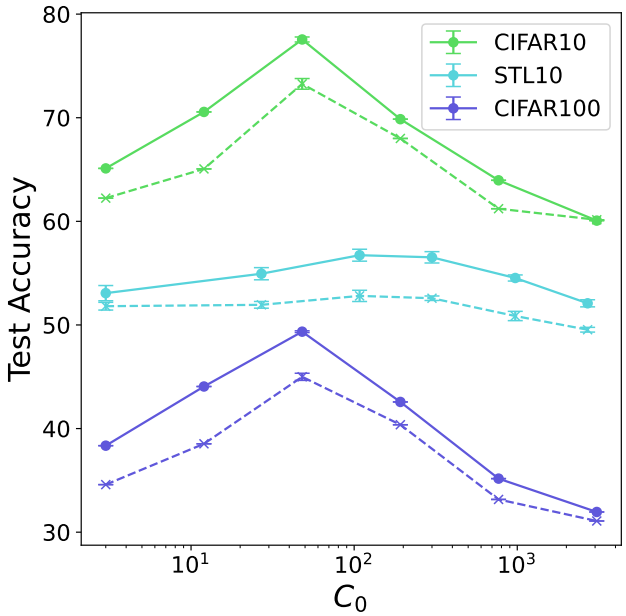

Figure S.5 . Dependence on C_0

layers have almost the same inductive bias, especially, in the upstream layers.

We utilized Tesla V100 GPUs and approximately 10 GPU hours for our experiments. Consider the S-Mixer architecture consisting of four blocks and $S = C = 64$. In this study, we trained a modified version of the S-Mixer architecture by replacing one of the four blocks with a block that incorporates random permutation. The training was conducted on CIFAR-10, CIFAR-100, and STL-10 datasets. The optimizer and training settings used in this experiment were consistent with those described in Section C.6.

D.6. Remark on input patch size

In this study, we focused on changing the size of the mixing layers and fixed the input token and channel size (S_0, C_0). In other words, the patch size P, satisfying $C_0 = 3P^2$, is fixed. While we observe that our experimental results hold regardless of the patch size, one naive question is whether there is any optimal patch size for achieving the highest accuracy. Although this is beyond the scope of this study, we show the performance depending on C_0 in Figure S.5 as a side note. The number of mixing layers is fixed at $C = S = 64$. We observed that the optimal C_0 depended on data; $C_0 = 48$ $(S_0 = 64)$ for CIFAR-10 and 100, and $C_0 = 108$ $(S_0 = 225)$ for STL-10. Note that the dimension of an input image is $S_0C_0 = 3,072$ for CIFAR datasets and $24,300$ for STL-10. It would be rational that the optimal patch size depends on the detailed information of data.

We utilized Tesla V100 GPUs and approximately 30 GPU hours for our experiments. We conducted training experiments on